

Software Requirements Specification

Public Representative & Civic Works Transparency Platform

Version 1.0

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Prepared for: JanDrishti Project Team

Derived from: JanDrishti Business Requirements Specification (BRS) v2.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for JanDrishti, a public representative and civic-works transparency platform. It translates the goals and features described in the JanDrishti Business Requirements Specification (BRS) v2.0 into requirements precise enough to guide design, development, and testing. This document is intended for the development team, QA team, project sponsors, and technical reviewers.

1.2 Scope

JanDrishti will allow elected representatives (MLA, MP, MLC, and local body members) to publish promises, projects, budgets, timelines, and evidence for their constituency. Citizens will be able to view, verify, rate, and raise grievances against this published information through a web application, mobile-responsive site, WhatsApp chatbot, and SMS/IVR channel. The system will also expose a public read-only API for researchers and civil-society organisations.

Out of scope for the initial releases: direct financial disbursement to contractors, automated legal or disciplinary action, and real-time video surveillance of project sites.

1.3 Definitions, Acronyms and Abbreviations

Term Definition

BRS Business Requirements Specification --- the source document this SRS is derived from

SRS Software Requirements Specification --- this document

MLA / MP / MLC Member of Legislative Assembly / Parliament / Legislative Council

FUC Fund Utilization Certificate --- an official document certifying how allocated funds were spent

SLA Service Level Agreement --- the maximum time allowed to resolve a grievance

IVR Interactive Voice Response --- automated telephone menu system

API Application Programming Interface

RBAC Role-Based Access Control

DPDP Act Digital Personal Data Protection Act, 2023 (India)

REQ-F-xxx Unique identifier for a functional requirement in this document

REQ-N-xxx Unique identifier for a non-functional requirement in this document

1.4 References

- JanDrishti Business Requirements Specification (BRS), Version 2.0, 25 July 2026
- IEEE Std 830-1998 --- Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018 --- Systems and software engineering --- Life cycle processes --- Requirements engineering
- Digital Personal Data Protection Act, 2023 (India)

1.5 Document Overview

Section 2 describes the product at a high level. Section 3 lists detailed functional requirements grouped by feature area. Section 4 defines external interface requirements. Section 5 defines non-functional requirements. Section 6 gives a high-level architecture view. Section 7 describes core data entities. Section 8 documents representative use cases. Section 9 lists other requirements including legal and compliance items. Section 10 provides a traceability matrix back to the BRS, and Section 11 is the appendix.

2. Overall Description

2.1 Product Perspective

JanDrishti is a new, standalone, cloud-hosted system. It integrates with existing government systems (e-tendering portals, open data portals) as external data sources

rather than replacing them. It also integrates with third-party communication channels: WhatsApp Business API, an SMS/IVR gateway, and push notification services.

2.2 Product Functions (Summary)

- Publish and track election promises through their full lifecycle.
- Track projects, budgets, and timelines per constituency and per village/ward.
- Store and verify evidence (photos, videos, documents) with a tamper-evident log.
- Collect citizen feedback and manage grievances against specific projects with SLA tracking.
- Support social audits by local citizen verification committees.
- Provide multi-channel access: web, mobile-responsive app, WhatsApp chatbot, SMS/IVR.
- Generate analytics: AI delay prediction, transparency score, constituency development index.
- Expose a public, rate-limited read-only API for external analysis.

2.3 User Classes and Characteristics

User Class Characteristics

Citizen General public; ranges from smartphone-literate urban users to low-literacy rural users on feature phones; primary need is simple, trustworthy information

Representative / Publishes and maintains promises, projects, and Representative's evidence on behalf of a representative; moderate Office Staff technical proficiency assumed

Department Official Confirms budget and project data; accesses the system occasionally, typically via desktop

Verification Local citizen volunteer who co-signs social Committee Member audits; app or web access, moderate technical proficiency

Platform Manages accounts, roles, moderation, and Administrator fraud/anomaly review; high technical proficiency

External Developer / Consumes the public API; technical, works outside
Researcher the JanDrishti UI entirely

2.4 Operating Environment

- Server-side: cloud infrastructure (AWS or GCP), containerised services, auto-scaling.
- Client-side (web): modern evergreen browsers (Chrome, Firefox, Safari, Edge) on desktop and mobile.
- Client-side (mobile): responsive web app; native app considered from Phase 3 onward.

- Low-connectivity access: offline-capable mode with local storage and periodic sync; SMS/IVR fallback for feature phones.

2.5 Design and Implementation Constraints

- Must support a minimum of five regional languages at launch.
- Must use TLS 1.3 for all data in transit and AES-256 for data at rest.
- Must expose an append-only, hash-chained log for all evidence and status-change events.
- Must comply with the Digital Personal Data Protection Act, 2023, for any personal data collected from citizens.

2.6 Assumptions and Dependencies

- Government e-tendering and open data portals expose stable, accessible interfaces for read-only integration.
- A WhatsApp Business API account and an SMS/IVR gateway account are provisioned before Phase 4.
- Representatives' offices are willing to regularly enter and update promise/project data.

3. System Features and Functional Requirements

Each functional requirement has a unique ID, a testable requirement statement, and a priority (High / Medium / Low). Requirement IDs are stable identifiers to be used in design documents, test cases, and the traceability matrix in Section 10.

3.1 User Management and Access Control

ID Requirement Priority

REQ-F-001 The system shall allow a user to register and High authenticate using a mobile number and OTP, or email and password.

REQ-F-002 The system shall support four roles at minimum: High Administrator, Representative (and delegated staff), Department Official, and Citizen.

REQ-F-003 The system shall enforce role-based access High control (RBAC) so that a user can only perform actions permitted for their role.

REQ-F-004 The system shall allow an Administrator to Medium suspend, verify, or reassign any user account.

3.2 Promise Tracking

ID Requirement Priority

REQ-F-005 The system shall allow a Representative to High create a promise record with a title, category, source (manifesto/speech/other), and target date.

REQ-F-006 The system shall allow a promise to be set to High one of the following statuses: Committed, In Progress, Completed.

REQ-F-007 The system shall allow a Citizen to search and High filter promises by keyword, category, village/ward, and status.

REQ-F-008 The system shall log every status change to a High promise with a timestamp and the user who made the change.

3.3 Project and Budget Management

ID Requirement Priority

REQ-F-009 The system shall allow a Representative or High Department Official to create a project, optionally linked to a promise.

REQ-F-010 The system shall record a project's expected High start date, expected completion date, and current status (Completed / Ongoing / Pending).

REQ-F-011 The system shall record budget allocated, High budget spent, and budget remaining for each project, broken down by sector.

REQ-F-012 The system shall display an aggregate project High dashboard per constituency, filterable by status.

REQ-F-013 The system shall display a village/ward-level High dashboard listing all works with status and budget details.

REQ-F-014 The system shall flag a project as delayed if Medium its current date exceeds its expected completion date while status is not Completed.

3.4 Evidence and Verification

ID Requirement Priority

REQ-F-015 The system shall allow a Representative or High Department Official to upload photo, video, and PDF evidence against a project.

REQ-F-016 The system shall compute and store a High cryptographic hash for every evidence file at upload time, chained to the previous entry in an append-only log.

REQ-F-017 The system shall allow a Representative to Medium upload a Fund Utilization Certificate (FUC) against a project's budget line.

REQ-F-018 The system shall flag a discrepancy where the Medium claimed spend for a project differs from the amount stated in its uploaded FUC by more than a configurable threshold.

REQ-F-019 The system shall allow designated Verification Medium Committee members to co-sign a project as socially audited, recording their identity and timestamp.

3.5 Citizen Engagement

ID Requirement Priority

REQ-F-020 The system shall allow a Citizen to submit a High rating, comment, or suggestion against a specific project.

REQ-F-021 The system shall allow a Citizen to submit a High grievance against a specific project, with an option to remain anonymous where legally permitted.

REQ-F-022 The system shall assign an SLA deadline to High every submitted grievance and route it to the

responsible Department Official.

REQ-F-023 The system shall allow a grievance to be Medium escalated automatically if unresolved after its SLA deadline.

REQ-F-024 The system shall send a notification to a High subscribed Citizen when a followed project's status changes.

REQ-F-025 The system shall send an opt-in weekly digest Low (SMS, email, or WhatsApp) summarising changes in a citizen's selected village/ward.

3.6 Multi-Channel Access

ID Requirement Priority

REQ-F-026 The system shall provide a WhatsApp chatbot Medium that accepts a project name or ID and returns its current status and a link to its detail page.

REQ-F-027 The system shall provide an SMS keyword and IVR Medium short code through which a citizen can retrieve a project's status without a smartphone or data plan.

REQ-F-028 The system shall support voice-based search for Low project and promise lookup in at least one regional language at launch.

REQ-F-029 The system shall support a minimum of five High regional languages for all citizen-facing UI text at launch.

REQ-F-030 The system shall function in an offline-capable Medium mode, storing entries locally and syncing automatically once connectivity is available.

3.7 Analytics and Intelligence

ID Requirement Priority

REQ-F-031 The system shall generate a delay-risk Medium indicator (Low/Moderate/High) for each ongoing project using historical timeline data.

REQ-F-032 The system shall compute a Transparency Score Medium per representative based on completeness of promise, evidence, and update records.

REQ-F-033 The system shall compute a Constituency Medium Development Index from completion rate and spend efficiency, and shall publish the calculation method alongside the score.

REQ-F-034 The system shall auto-generate a monthly Low performance report per representative in PDF format.

REQ-F-035 The system shall generate an anonymised, Low geo-clustered heatmap of grievance volume.

3.8 Integrations and Open Data

ID Requirement Priority

REQ-F-036 The system shall generate a unique, scannable High QR code for every project, linking to its public detail page.

REQ-F-037 The system shall periodically pull tender Low status from connected government e-tendering portals where an integration exists.

REQ-F-038 The system shall periodically cross-check Low budget figures against public data available on connected open-data portals (e.g. data.gov.in) where an integration exists.

REQ-F-039 The system shall expose a read-only, Medium rate-limited public API returning anonymised promise, project, and budget data in JSON.

REQ-F-040 The system shall provide a before/after image Low comparison slider for any project with both 'before' and 'after' evidence uploaded.

4. External Interface Requirements

4.1 User Interfaces

- A public web dashboard (desktop and mobile-responsive) for citizens, requiring no login for read-only browsing.

- An authenticated portal for Representatives, staff, and Department Officials to enter and manage data.
- An administrative console for platform Administrators, with audit-log and moderation views.

4.2 Hardware Interfaces

- No dedicated hardware is required. Feature-phone users interact only through standard SMS/IVR telephony hardware.

4.3 Software Interfaces

Interface Purpose

Government e-Tendering Read-only pull of live tender status for Portal API linked projects

Open Government Data Read-only cross-verification of published Portal (e.g. budget figures data.gov.in)

WhatsApp Business API Chatbot-based citizen status queries

SMS / IVR Gateway Text and voice-based status queries for feature-phone users

Cloud Object Storage Storage of evidence files (photos, videos, (e.g. S3/GCS) PDFs)

Public REST API Read-only access for external developers, journalists, and researchers

4.4 Communication Interfaces

- All client-server communication shall use HTTPS (TLS 1.3).
- Push notifications shall be delivered via a standard mobile push notification service (e.g. FCM/APNs equivalent for the web app).

5. Non-Functional Requirements

5.1 Performance

- REQ-N-001: The system shall load the public project dashboard within 2.5 seconds on a 3G connection.
- REQ-N-002: Cached read API responses shall return within 500 ms under normal load.

5.2 Scalability

- REQ-N-003: The system shall support at least 1,000,000 registered citizen accounts and 500 representative accounts without architectural redesign.

5.3 Availability and Reliability

- REQ-N-004: The public-facing site shall maintain 99.5% uptime, measured monthly.
- REQ-N-005: The system shall support an offline-first mode for village-level data entry with automatic sync on reconnection.

5.4 Security

- REQ-N-006: All data at rest shall be encrypted using AES-256.
- REQ-N-007: All data in transit shall be encrypted using TLS 1.3.
- REQ-N-008: The system shall maintain an append-only audit log of every data change, including actor, timestamp, and before/after values.
- REQ-N-009: The public API shall enforce rate limiting to prevent scraping abuse.

5.5 Usability and Accessibility

- REQ-N-010: Citizen-facing content shall be written at approximately a Class 8 reading level.
- REQ-N-011: The system shall support screen readers, keyboard navigation, and high-contrast display modes.
- REQ-N-012: The system shall provide a dark mode display option.

5.6 Localization

- REQ-N-013: The system shall support a minimum of five regional languages at launch, with an extensible translation framework for adding more.

5.7 Maintainability

- REQ-N-014: The system shall be built as modular services so that individual feature areas (e.g. grievances, evidence, analytics) can be updated independently.

5.8 Legal and Compliance

- REQ-N-015: The system shall comply with the Digital Personal Data Protection Act, 2023, for any personal data collected from citizens.
- REQ-N-016: The system shall allow citizens to submit feedback and grievances anonymously where legally permitted.

6. High-Level Architecture Overview

JanDrishti follows a layered architecture:

- Presentation Layer --- responsive web application, WhatsApp chatbot interface, and SMS/IVR gateway integration.
- API Layer --- a secured REST API gateway handling authentication, rate limiting, and routing to backend services; also exposes the public read-only API.

- Service Layer --- modular services for Promise Tracking, Project & Budget Management, Evidence & Verification, Grievance Management, Notifications, and Analytics.
- Data Layer --- a relational database for structured records (promises, projects, budgets, grievances) and object storage for evidence files, with an append-only hash-chained log for tamper-evidence.
- Integration Layer --- connectors to government e-tendering portals, open data portals, WhatsApp Business API, and SMS/IVR providers.

Deployment shall use containerised services on a scalable cloud platform (AWS or GCP) with auto-scaling and load balancing across the API and Service layers.

7. Data Requirements

The core entities and their key attributes are summarised below. A representative may have many promises; a promise may relate to zero or one projects directly (some projects exist independent of a specific promise); a project has one budget breakdown, one or more evidence items, and zero or more grievances and feedback entries.

Entity Key Attributes

Representative id, name, tier (MLA/MP/MLC/Local), constituency, party, term dates

Promise id, representative_id, title, category, source, status

Project id, promise_id (nullable), village/ward_id, budget_id, timeline, status, evidence[]

Budget Line id, project_id, sector, allocated, spent, remaining, FUC_reference

Evidence id, project_id, type, hash, previous_hash, uploaded_by, timestamp

Grievance id, project_id, citizen_id (nullable if anonymous), description, SLA_deadline, status

Village/Ward id, name, district, population, constituency_id

Citizen Feedback id, project_id, rating, comment, verified_flag

Social Audit id, project_id, committee_member_id, verified_at, Record remarks

8. Use Case Descriptions

8.1 UC-01: Submit and Track a Grievance

Actor: Citizen

Precondition: The citizen has located a project on the platform

Main Flow: 1) Citizen selects a project. 2) Citizen submits a grievance with a description, optionally anonymously. 3) System assigns an SLA deadline and routes the grievance to the responsible Department Official. 4) Citizen receives a tracking reference and can view status updates.

Postcondition: The grievance is recorded, assigned, and visible to the citizen with a live status

Alternate Flow: If the SLA deadline passes without resolution, the system automatically escalates the grievance

8.2 UC-02: Verify a Completed Project (Social Audit)

Actor: Verification Committee Member

Precondition: A project has been marked Completed with evidence uploaded

Main Flow: 1) Committee member reviews the project's evidence. 2) Committee member records a co-signed verification, with remarks if needed. 3) System stores the social audit record and updates the project's verification status.

Postcondition: The project displays a 'Socially Verified' indicator visible to citizens

8.3 UC-03: Publish a New Promise and Link a Project

Actor: Representative / Representative's Office Staff

Precondition: The representative account is authenticated

Main Flow: 1) Representative creates a promise record (title, category, source, target date). 2) Representative creates one or more linked projects with budget and timeline. 3) Representative uploads initial evidence, if available. 4) System publishes the promise and linked project(s) to the public dashboard.

Postcondition: The promise and project are visible and searchable by citizens

8.4 UC-04: Query Project Status via SMS

Actor: Citizen (feature-phone user)

Precondition: Citizen knows the project name or a shared keyword

Main Flow: 1) Citizen sends an SMS keyword with the project name to the published short code. 2) System matches the project and replies with its current status and a shortened link. 3) Citizen may reply to subscribe to future status-change alerts by SMS.

Postcondition: The citizen receives the project status without needing a smartphone or data connection

9. Other Requirements

9.1 Legal and Regulatory

- The system shall retain grievance and audit records for a minimum period consistent with applicable government record-keeping norms.
- The system shall provide a mechanism for citizens to request correction or removal of their personal data in line with the DPDP Act, 2023.

9.2 Auditability

- The system shall make its administrative and evidence audit logs available to authorised oversight bodies on request.

10. Traceability Matrix (BRS to SRS)

This matrix maps each major BRS section to the SRS requirements that implement it, to support review and change-impact analysis.

BRS Section SRS Requirement IDs

6. Critical Features --- Phase REQ-F-001 to REQ-F-014 1

7. Advanced Features REQ-F-031, REQ-F-034, REQ-F-036, (existing) REQ-F-037, REQ-F-040, REQ-F-026, REQ-F-028, REQ-F-029

7.1 New Modules (Grievance, REQ-F-015 to REQ-F-025, REQ-F-032, Social Audit, FUC, Multi-tier, REQ-F-033, REQ-F-035, REQ-F-038, Manifesto, Tamper-Evident Log, REQ-F-039 SMS/IVR, Dev Index, Open Data, Heatmap, Digest, Public API)

8. Non-Functional Requirements REQ-N-001 to REQ-N-016

9. System Architecture & Section 6 (Architecture Overview), Security REQ-N-006 to REQ-N-009

10. Core Data Entities Section 7 (Data Requirements)

14. Risks & Mitigations REQ-F-016 to REQ-F-019, REQ-F-030, REQ-N-009

11. Appendix

11.1 Requirement Priority Definitions

Priority Definition

High Required for the Phase 1 (MVP) release; the platform cannot launch without it

Medium Required within the first two to three releases; important for adoption and trust

Low Enhances the platform but can be deferred to a later phase without blocking core value

11.2 Open Items for Stakeholder Confirmation

- Final list of the five (or more) regional languages to support at launch.
- Selection of SMS/IVR and WhatsApp Business API vendors.
- Governance process for appointing local Verification Committee members.
- Configurable threshold value for flagging FUC/spend discrepancies.

This SRS is derived from and should be read alongside the JanDrishti Business Requirements Specification (BRS) v2.0.